What are IP, MAC, DNS, DHCP and ports? Include a table of 12 common ports: 21, 22, 23, 25, 53, 80, 110, 143, 443, 445, 3306, 3389 with service and typical risk.

IP, MAC, DNS, DHCP and Ports

A. IP Address (Internet Protocol Address)

An IP address is a logical numerical address assigned to a network interface to identify and communicate with devices across IP networks.

- **IPv4 example**: 192.168.1.10
- **IPv6 example**: 2001:db8::1

IP addresses help route packets between networks.

B. MAC Address (Media Access Control Address)

A MAC address is a link-layer address associated with a network interface, commonly represented as a 48-bit hexadecimal value.

Example: 00:1A:2B:3C:4D:5E

It is used to deliver frames within a local network. MAC addresses can also be changed or spoofed.

C. DNS (Domain Name System)

DNS translates human-readable domain names into IP addresses and provides other types of domain information.

Example: When you enter www.google.com, DNS helps your computer find an IP address for the website.

D. DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns network configuration to devices, such as:

- IP address
- Subnet mask
- Default gateway
- DNS server information

Example: When your laptop connects to a Wi-Fi router, DHCP usually provides its local IP configuration.

E. Ports

A port is a logical communication endpoint identified by a 16-bit number in TCP or UDP.

Ports allow multiple network services to operate on a single IP address.

Example: A web server commonly uses TCP port 443 for HTTPS traffic.

12 Common Networking Ports

Port	Service	Typical security risk
21/TCP	FTP	Cleartext credentials; unauthorized file access
22/TCP	SSH	Brute-force attacks; stolen credentials
23/TCP	Telnet	Unencrypted login credentials
25/TCP	SMTP	Open relay abuse; email spoofing or spam
53/TCP, UDP	DNS	DNS cache poisoning; amplification attacks
80/TCP	HTTP	Unencrypted traffic; web vulnerabilities
110/TCP	POP3	Cleartext credentials without TLS
143/TCP	IMAP	Credential interception without TLS
443/TCP	HTTPS	Web application flaws; TLS misconfiguration
445/TCP	SMB	Unauthorized file access; worm propagation
3306/TCP	MySQL	Unauthorized database access; weak authentication
3389/TCP, UDP	RDP	Brute-force attacks; exposed remote access

Important - An open port does not automatically mean a system is vulnerable. Risk depends on the running service, configuration, patch level, and access controls.